

Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract

Low-resource regional sub-dialects face severe data scarcity and lack parallel text normalization benchmarks. In this paper, we address dialect normalization for Marathi—converting non-standard text from three major regional dialects (Southern Konkan, Northern Konkan, and Varhadi) into Standard Written Marathi. To overcome data scarcity without expensive human annotation, we introduce an automated synthetic parallel expansion pipeline combining Gemma-4 translation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), expanding the clean parallel corpus from 16,163 to 32,335 verified pairs. We fine-tune sequence-to-sequence transformer architectures (mT5-small and IndicBART) using task-specific normalization conditioning tags under a 5-fold cross-validation scheme. Empirical results show that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the expanded dataset (+12.39 BLEU over IndicBART). Crucially, multi-tier verification prevents synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion eliminates cross-dialect parameter competition.

Keywords (at least 4-5): Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modeling, Synthetic Data Augmentation, Task-Specific Conditioning, mT5, IndicBART.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] has made it possible to develop systems capable of understanding and generating natural language. Open-weight models have gained popularity to process text in various domains as they provide a finer control over system architecture, runtime environment, and training data [2]. Yet current language systems tend to work efficiently with standard high-resource languages. Queries based on non-standard input yield unexpected and undesired outputs. [3]. Speakers of regional dialects face many issues in systems that use these models across automated speech recognition, machine translation and other language processing systems [4, 5].

Dialect normalization is the task of translating non-standard regional dialects into standard written language. The main challenge to be addressed while designing such systems is the preservation of the meaning of original text. [6]. It serves to be an essential layer for digital linguistic inclusion. Countries having large and diverse speech communities, such as India, which has 22 official languages and hundreds of regional dialects spoken by 1.4 billion people, regional varieties have remained unattended in computational linguistic research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [7].

Development reliable pipelines for regional Indic dialects has faced these fundamental challenges:

1. **Resource Scarcity:** Parallel corpora mapping the regional dialects to standard written language do not exist. Curation of such sets by experts fluent in multiple dialects is both

expensive and unscalable [6, 8].

2. **Morphological and Dialectal Heterogeneity:** Marathi exhibits rich morphology where different words undergo varied complex transformations across different regions where the dialects are spoken [7, 9]. Consequently, evaluations vary significantly across individual dialects—ranging from systematic suffix alternation in Varhadi to intense lexical and phonetic divergence in Northern and Southern Konkan [7, 10].
3. **Underperformance of Multilingual Models:** Performance of pre-trained models like IndicBART [11], IndicNLP Suite [12], and mT5 [13] seems to drop drastically on non-standard language variants due to subword fragmentation during tokenisation and out-of-vocabulary phonological shifts [3, 14].

Our work mainly tries to investigate and conclude 1. how do normalisation scores vary across regional Marathi dialects which have different degrees of variations as compared to standard Marathi; 2. how effectively can existing sequence-to-sequence architectures adapt to dialect-to-standard mapping under task-specific conditioning; and 3. whether automated synthetic data expansion can accurately solve the problem of data scarcity and eliminate cross-dialect hallucinations and grammatical errors [15, 16]. To address these questions, we construct an end-to-end normalization framework combining LLM-based synthetic generation, multi-tier syntactic and semantic verification, and conditioned transformer fine-tuning across both single-dialect and multi-dialect settings.

Our core contributions are:

- **Parallel Corpus Generation:** Based on transcripts from the RESPIN dataset [17], we construct a parallel corpus for three of the Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi.
- **Automated Synthetic Augmentation with Verification:** We design a data expansion pipeline paired with a multi-tier verification engine, which has rule-based script filtration and LLM-based semantic inspection.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects [18] which were later replaced by pre-trained language models [19, 20]. Similar approaches have been developed for Finnish, Swedish, and Estonian dialects [21–23]. A study of four language families by Kuparinen et al. [24] suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules [25], while Mandarin scripts relied on acoustic-phonetic mapping [26]. The Conventional Orthography for Dialectal Arabic (CODA) framework [27] in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to multi-city translation benchmarks (MADAR [28], NADI [29]) and morphology-oriented prompt design for LLMs [30, 31].

2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART [11] and mT5 [13] which are trained on multilingual corpora cover major languages, they lack understanding of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP [32] offer pre-trained Marathi models for classification, but do not support generative text standardization. INDIC-DIALECT [14] considers Hindi varieties (Bhojpuri, Magahi, Maithili), for evaluation via WFST frameworks for text normalization while preserving the underlying semantics [33]. However, Marathi dialects were primarily dealt for acoustic/ASR classification [34, 35], or zero-shot LLM retrieval [36, 37].

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkani (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkani (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	All Dialects	809,934 (100%)	1,026.06	2,644	23,069

Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks [38–40]. Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model [15, 41, 42]. But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise [15, 16]. We solve this by utilising Gemma-4 with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

3 Methodology

3.1 Target Dialects and Linguistic Properties

We target the following major Marathi dialects that span the primary regions of Maharashtra and exhibit distinct linguistic and phonetic patterns based on the RESPIN corpus:

1. **Southern Konkani (Konkan Coast):** Characterized by shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Lexical differences include systematic vowel shifts: standard जातो (“he goes”) becomes Southern Konkani जातों; standard कुठे (“where”) becomes कुठेंय. Domain-specific terminology generally remains untouched but often accompanied by Konkani words.
2. **Northern Konkani (Khandesh Region):** Heavily influenced by neighboring Gujarati and Bhili language families, thus it exhibits significant consonant simplification and dental consonant shifts. Notable transformations include: standard कशाळा (“why”) → Northern Konkani कसाले; standard मुलगा (“boy”) → Northern Konkani पोर. Northern Konkani possesses the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (Vidarbha/Eastern Maharashtra):** Spoken in Vidarbha, it exhibits characteristic differences and verb suffix modifications. Standard interrogative का? (“why?”) becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains closest structural similarity to Standard Marathi as transformations are more systematic.

3.2 Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative [17], developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation.

The large speaker diversity ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus.

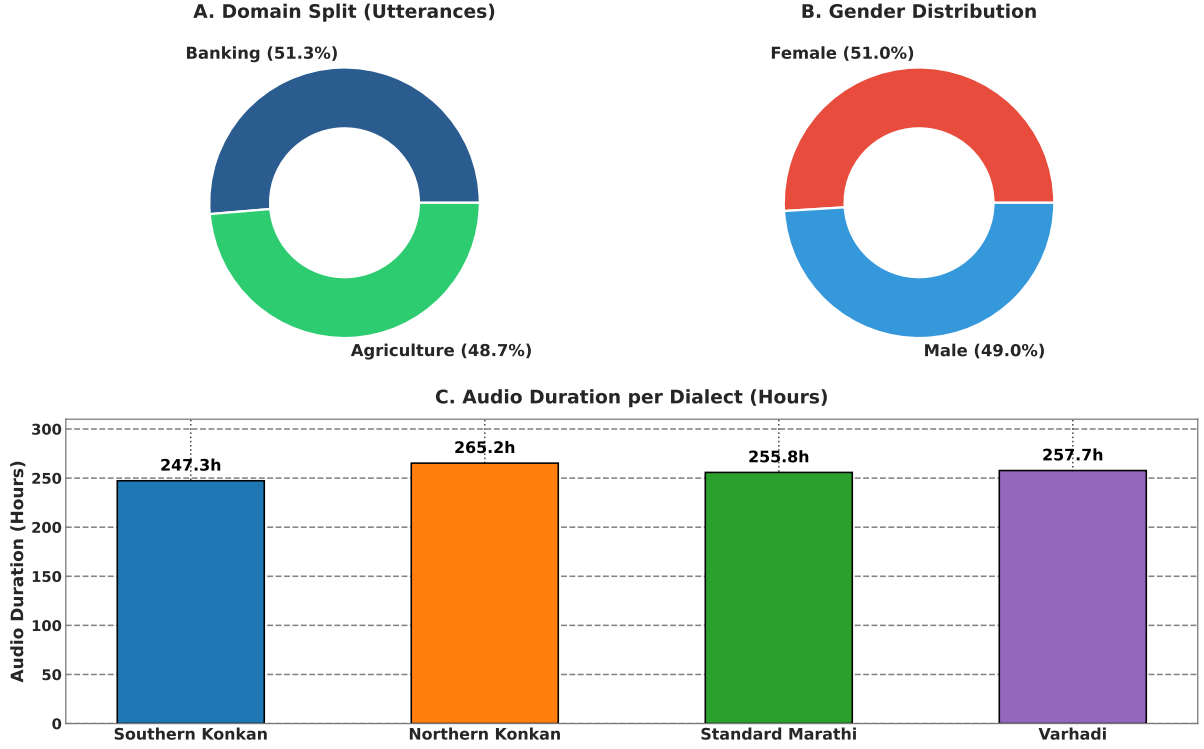


Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

3.3 LLM-Driven Translation and Verification Pipeline

To deal with data scarcity, we developed an automated synthetic data translation and generation pipeline powered by Gemma-4 [43] in 9B-instruction-tuned (gemma-4) configurations, accessed via Ollama. We considered several design decisions to ensure generation quality to be maximised:

- **1-Prompt-Per-Sentence Architecture:** Instead of generating batches of parallel pairs per API call (which risks array length mismatches and context contamination), each dialectal sentence is processed separately with a self-contained prompt.
- **Detailed Prompting Strategy:** The generation prompt enforces semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the input. The prompt includes dialect-specific lexical mapping tables and 5-8 few-shot examples per dialect.
- **Temperature Control:** Generation temperature is set to 0.3 with top- $p = 0.9$ to balance diversity with fidelity.

All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles [9], including:
 - *Devanagari Script Integrity:* Both source and target must possess $> 90\%$ Devanagari Unicode characters within the U+0900--U+097F range, automatically discarding unnecessary characters.
 - *Length Ratio Constraint:* $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
 - *Non-Identity Constraint:* $L_{\text{dialect}} \neq L_{\text{standard}}$ to exclude identical string copies.
 - *Minimum Token Count:* Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using llama-3.1-8b-instant (accessed via Groq API) performs semantic equivalence verification. The verifier rates

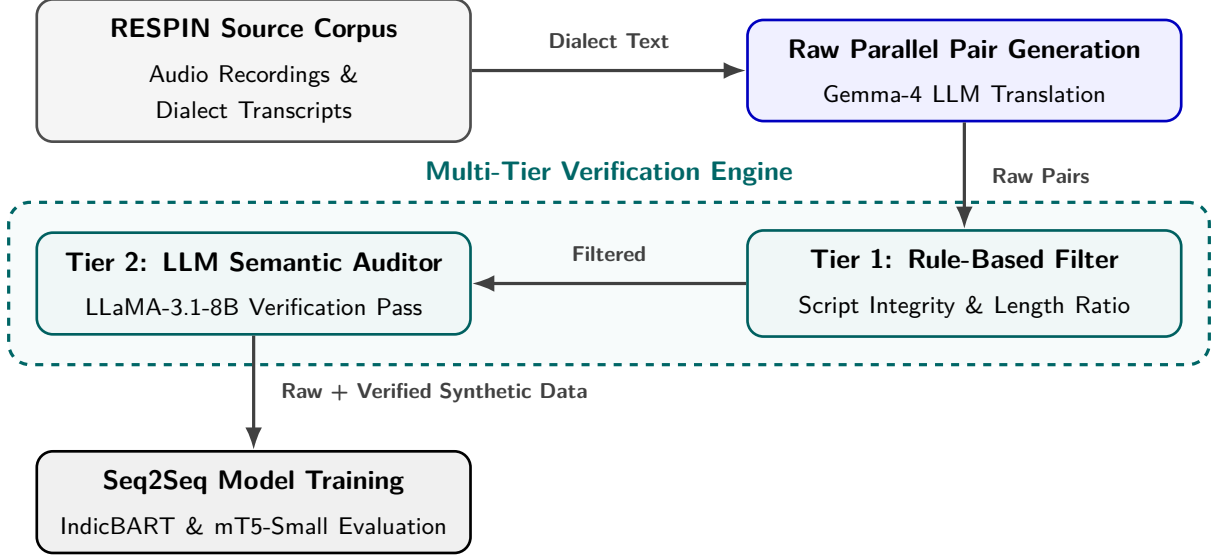


Figure 2: System architecture of the multi-dialect normalization framework.

Table 2: Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkani	5,576	5,838	5,569	269 (4.61%)	11,145
Northern Konkani	5,501	5,825	5,534	291 (5.00%)	11,035
Varhadi	5,086	5,330	5,069	261 (4.90%)	10,155
Total	16,163	16,993	16,172	821 (4.83%)	32,335

on a 1–5 score scale considering: (a) semantic preservation; (b) absence of hallucinated entities; (c) absence of Hindi/English contamination; and (d) complete standardization of dialects. Pairs scoring < 4.0 are rejected or routed to correction.

3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step correction loop: Step 1 generates corrected translations using specialised prompts based on the pre-defined reason of identified failure.

This leaves us with a total of 16,163 pairs

3.4 Synthetic Data Expansion

After construction the original 16,163 verified parallel pairs through the RESPIN corpus, we applied the same pipeline to synthetically expand it. The expansion phase uses a similar system prompt, inspired from that of translation phase to generate additional pairs from scratch that almost doubles the data size.

The pipeline was used to generate parallel pairs from the RESPIN transcripts and for synthetic data expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

3.5 Sequence-to-Sequence Model Conditioning and Fine-Tuning Setup

To convert the pre-trained seq2seq language architectures into dedicated dialect normalizers, models are fine-tuned under a 5-fold cross-validation scheme on an 85/15 stratified train-test split. Models were conditioned using specific tags attached before the actual text.

- **Task Normalization Prefixes:** A normalisation tag (`<normalize>`) is attached that instructs the encoder to perform the dialect-to-standard text transformation.
- **Dialect-Source Identification Tags:** Source sentences are prefixed with dialect markers

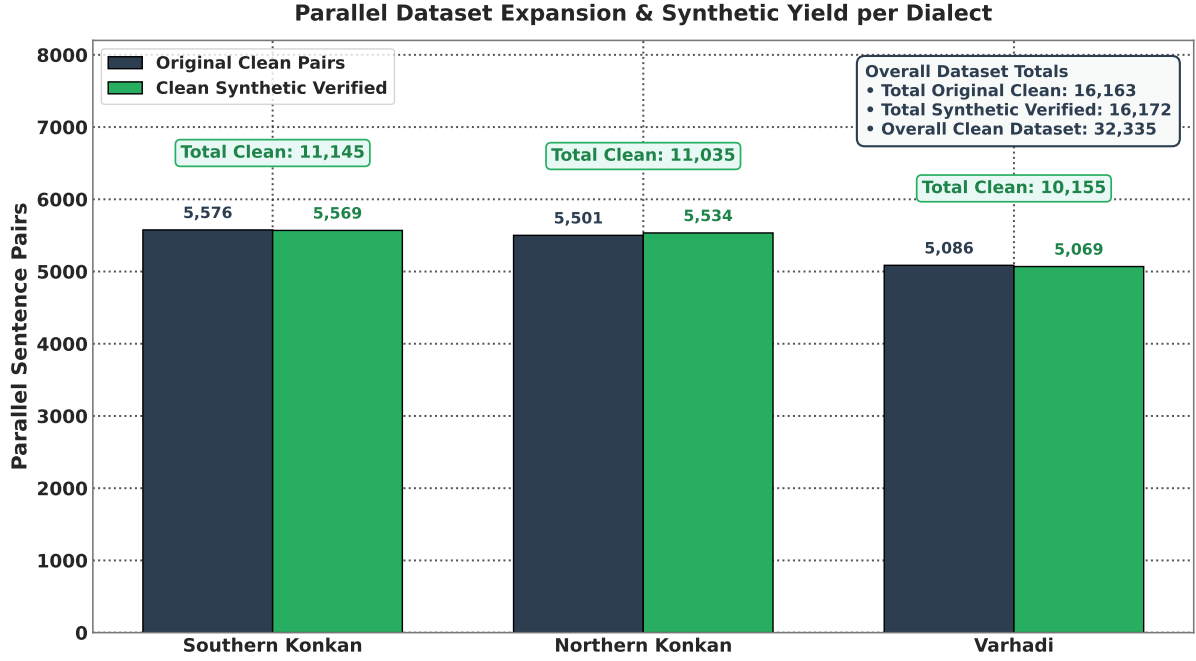


Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

Table 3: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required $\langle 2_{mr} \rangle$ token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

, $\langle D1 \rangle$ for Southern Konkan, $\langle D2 \rangle$ for Northern Konkan, and $\langle D4 \rangle$ for Varhadi for explicitly conditioning the model particularly in the multi-dialect training scenario.

- **Target Standardization Tags:** Decoder target conditioning is done using standard language tags ($\langle standard_{mr} \rangle$), enforcing the output alignment into formal Standard Written Marathi.

Table 3 details the underlying architectural properties and hyperparameter configurations for IndicBART (244M) and mT5-small (300M).

3.6 Evaluation Protocol

Even though we follow a standard train/test split strategy for stable performance evaluations, the final reports were generated on the official held-out test set provided by RESPIN which had around 2,170 utterances across target dialects after training the model on the complete train set. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions. Results are reported using metrics that are evaluated using SacreBLEU [44] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.

Table 4: Comprehensive Evaluation Matrix on IISc_RESPIN_test_mr Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkani	559	57.80	26.60%	43.86	34.37%
		Northern Konkani	540	90.13	6.46%	79.46	11.95%
		Varhadi	516	83.59	10.19%	74.81	14.73%
	Synthetically Expanded	Southern Konkani	559	24.06	60.32%	44.36	32.75%
		Northern Konkani	540	65.41	28.70%	79.76	12.61%
		Varhadi	516	67.97	25.43%	76.90	14.19%
Multi-Dialect	Original	Southern Konkani	559	26.08	52.98%	40.94	35.34%
		Northern Konkani	540	47.92	32.64%	77.50	14.65%
		Standard Benchmark	555	96.12	3.12%	96.65	1.91%
		Varhadi	516	73.04	26.96%	73.47	16.16%
		Overall Benchmark	2,170	62.98	30.13%	72.18	17.23%
	Synthetically Expanded	Southern Konkani	559	52.15	36.50%	43.88	34.96%
		Northern Konkani	540	54.35	25.65%	78.16	13.55%
		Standard Benchmark	555	96.80	1.65%	97.39	1.37%
		Varhadi	516	69.96	21.04%	74.74	15.57%
		Overall Benchmark	2,170	76.50	16.23%	73.48	16.58%

Table 5: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	5,576	837 / 836	48.54	78.35	46.31	74.43
Northern Konkani (D2)	5,501	826 / 825	60.58	80.30	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	81.00	91.21

- **Normalized WER / CER:** Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

4.1 Comprehensive Test Benchmark on IISc_RESPIN_test_mr

Key observations from Table 4 and Figure 4 include:

1. **mT5-Small Performance:** mT5-small upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling:** Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.

4.2 Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 5 presents the performance matrix on original single-dialect benchmarks, and Table 6 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 7 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 8 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

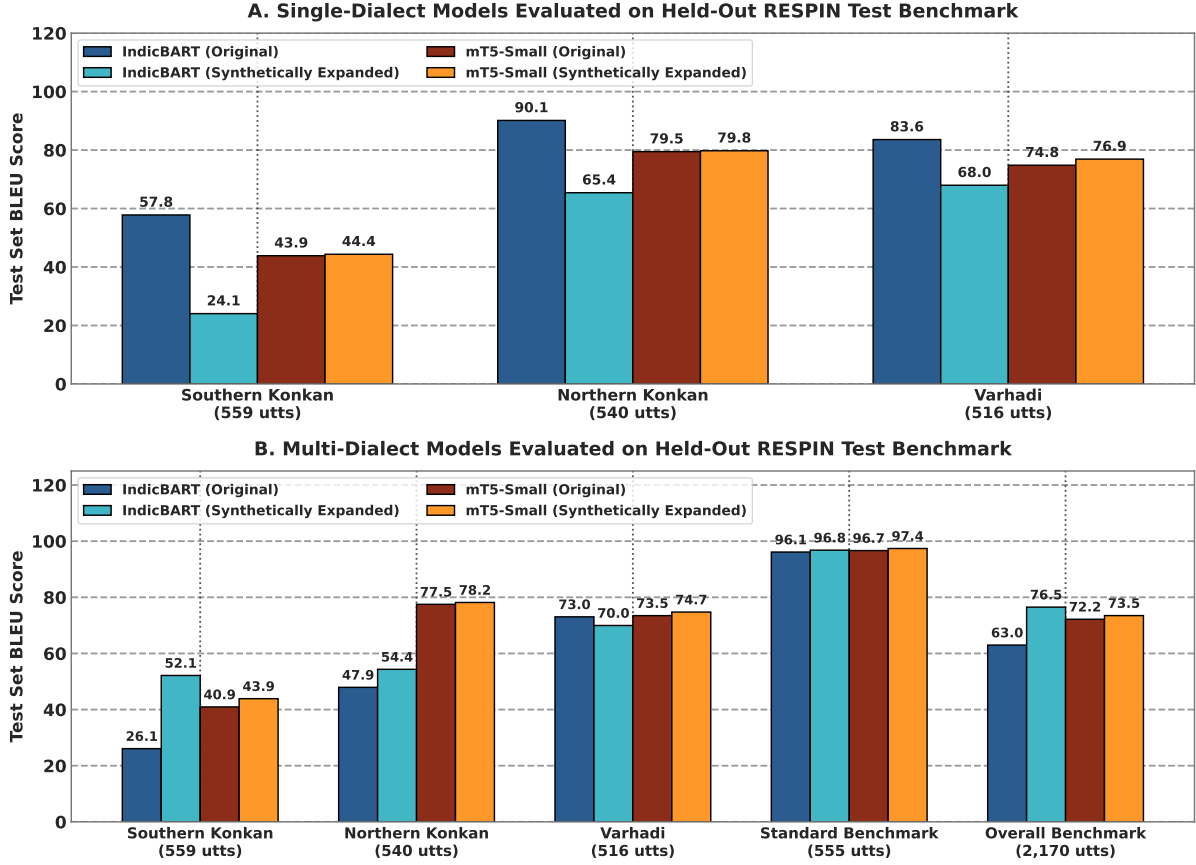


Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

4.3 Impact of Synthetic Data Augmentation

Table 9 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

Analyzing Table 9 through data availability and model architecture dimensions clarifies these dynamics:

- **Data Availability & Dialect Sparsity:** Southern Konkani (D1) suffered from extreme baseline data sparsity (26.21 BLEU for IndicBART, 48.28 BLEU for mT5-small). Doubling clean training pairs via synthetic expansion provided massive performance gains (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small) by supplying critical subword affix mappings. Conversely, Varhadi (D4) already possessed a saturated baseline (72.75–79.57 BLEU), where synthetic expansion caused a minor precision drop (−2.48 and −0.84 BLEU) due to LLM-generated suffix over-generalization slightly diluting exact N -gram matches against gold references.
- **Model Architecture Differences:** mT5-small consistently outperforms IndicBART across both original (63.29 vs. 48.17 BLEU) and expanded (69.51 vs. 57.12 BLEU) combined benchmarks. This architectural superiority stems from: (1) **Subword Vocabulary Capacity:** mT5-small’s 250,000-token SentencePiece vocabulary (mC4) segments non-standard dialect affixes with lower fragmentation than IndicBART’s 64,000 Devanagari-unified vocabulary; and (2) **Relative Positional Embeddings:** mT5’s relative positional bias models agglutinative dialectal suffix merges far more effectively than IndicBART’s absolute sinusoidal encodings.

Table 6: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	16,163	837 / 836	26.21	53.15	48.28	75.04
Northern Konkani (D2)	16,163	826 / 798	47.59	67.16	62.35	78.92
Varhadi (D4)	16,163	763 / 790	72.75	85.63	79.57	90.51
Overall Combined	16,163	2,426 / 2,424	48.17	68.58	63.29	81.37

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	11,145	1,672 / 1,671	47.25	64.69	65.10	82.88
Northern Konkani (D2)	11,035	1,656 / 1,655	40.51	62.21	62.07	79.29
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	78.89	90.59

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,742 rejected pairs that contained garbage translations. Table 10 presents the quantitative comparison on the benchmark suite.

Analyzing Table 10 through data quality and architectural resilience dimensions yields three key insights:

1. **Data Quality & Error Propagation:** Training on raw unverified synthetic data (19,914 pairs) causes severe performance degradation (-14.03 BLEU for IndicBART, -8.30 BLEU for mT5-small). Unfiltered LLM outputs contain hallucinated stems, script noise, and spurious suffix rules, which inject corrupted gradient signals during backpropagation. Multi-tier verification (rule-based script filtering and LLM semantic auditing) removes 3,742 flawed pairs while expanding clean data to 32,335 pairs, successfully recovering BLEU scores to 57.12 (IndicBART) and 69.51 (mT5-small).
2. **Architectural Resilience to Synthetic Noise:** IndicBART exhibits extreme sensitivity to synthetic noise, suffering a 24.6% relative BLEU drop on unverified data ($57.12 \rightarrow 43.09$). In contrast, mT5-small demonstrates superior structural resilience with a smaller relative drop of 11.9% ($69.51 \rightarrow 61.21$). This resilience is directly attributed to mT5’s larger 300M parameter capacity and extensive 101-language pre-training, which provides stronger regularization against noisy subword alignment gradients.
3. **Essential Role of Multi-Tier Auditing:** Combining rule-based orthographic checks with LLM-based semantic verification ensures that synthetic data augmentation scales quantitative performance without introducing systematic morphological drift.

4.5 Model Architecture Analysis

As shown in Table 3, the 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage:** mT5’s 250K vocabulary provides $3.9\times$ more subword units than IndicBART’s 64K, enabling finer-grained tokenization of morphologically complex Marathi

Table 8: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialect-wise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	32,335	1,672 / 1,710	52.06	72.15	66.62	83.57
Northern Konkani (D2)	32,335	1,656 / 1,627	49.08	69.90	62.72	79.63
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	78.73	90.46
Overall Combined	32,335	4,852 / 4,850	57.12	75.49	69.51	84.58

Table 9: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkani (D1)	26.21	52.06	+25.85	48.28	66.62	+18.34
Northern Konkani (D2)	47.59	49.08	+1.49	62.35	62.72	+0.37
Varhadi (D4)	72.75	70.27	-2.48	79.57	78.73	-0.84
Overall Combined	48.17	57.12	+8.95	63.29	69.51	+6.22

dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

2. **Positional Encoding:** mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead:** IndicBART requires explicit language tokens ($\langle 2_{mr} \rangle$) that add conditioning complexity, whereas mT5’s pure text-to-text format enables direct mapping without task-specific architectural constraints.

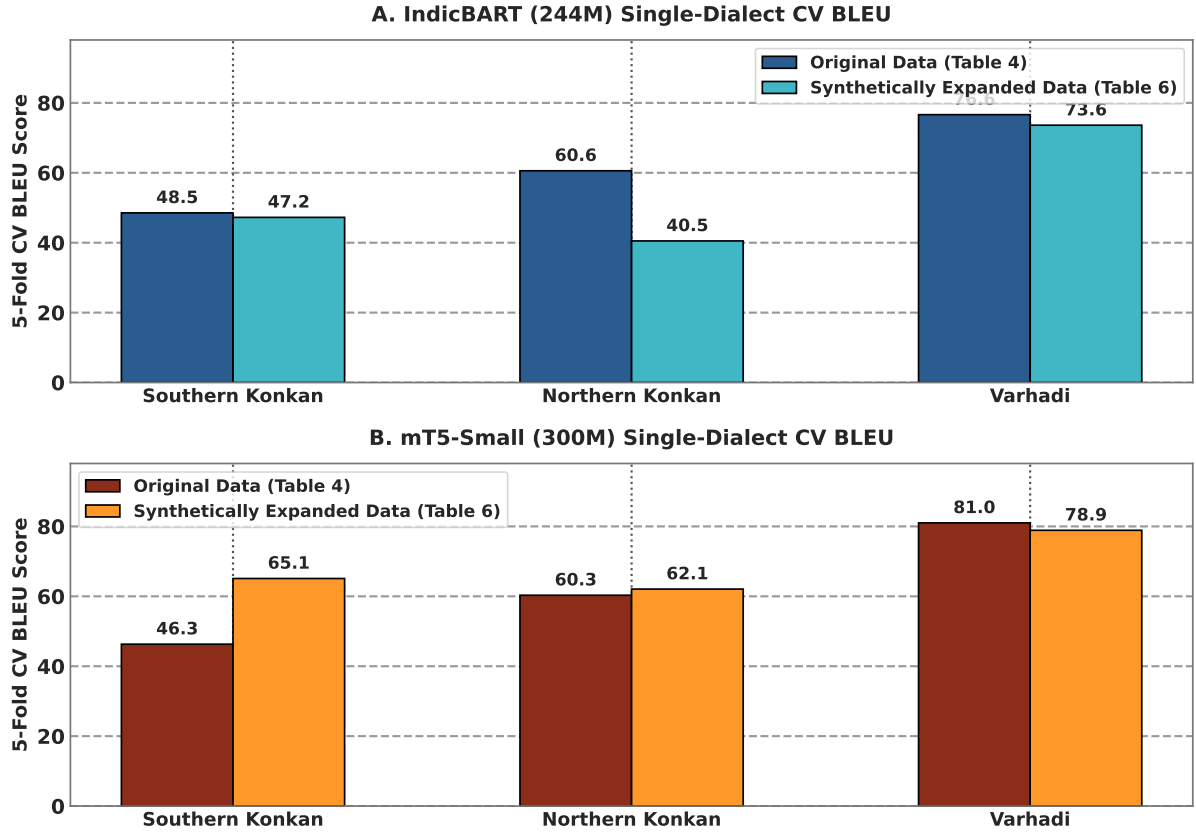


Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

Table 10: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	-14.03	-10.95
	Filtered Clean Data	32,335	57.12	75.49	+14.03	+10.95
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	-8.30	-4.40
	Filtered Clean Data	32,335	69.51	84.58	+8.30	+4.40

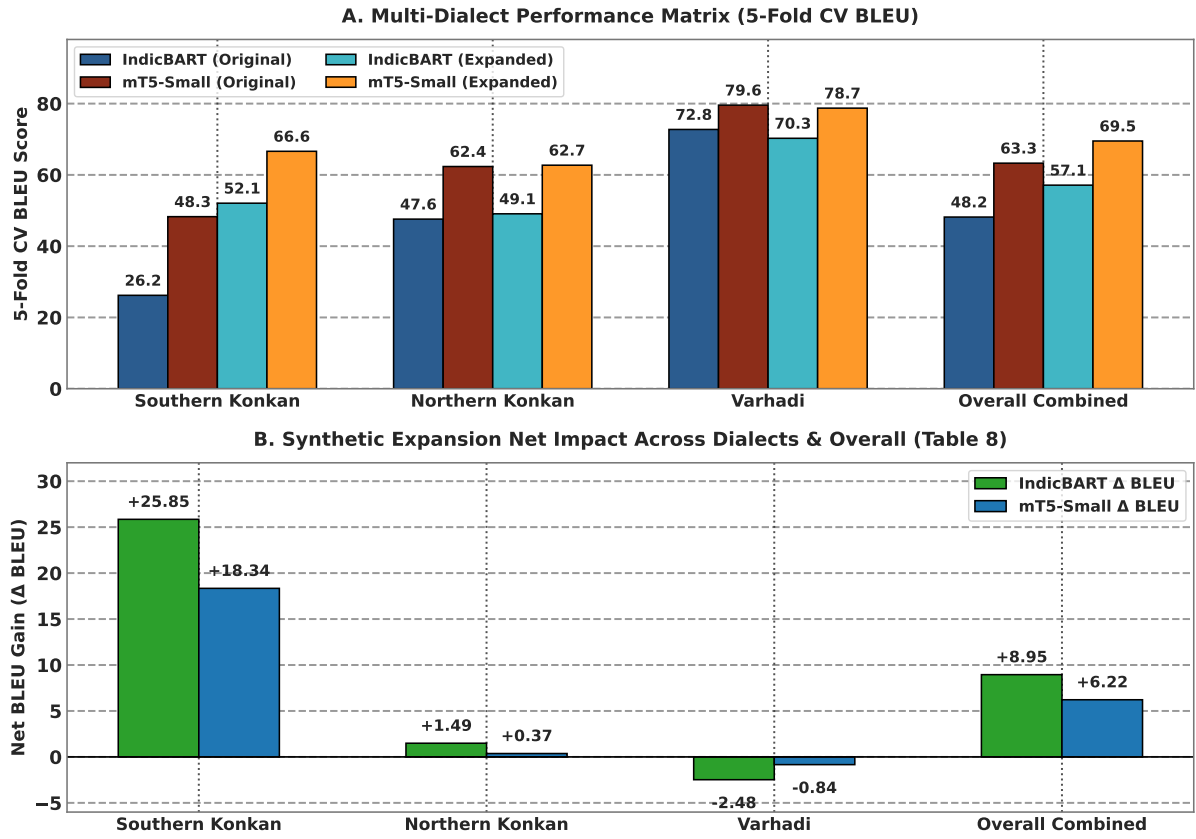


Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

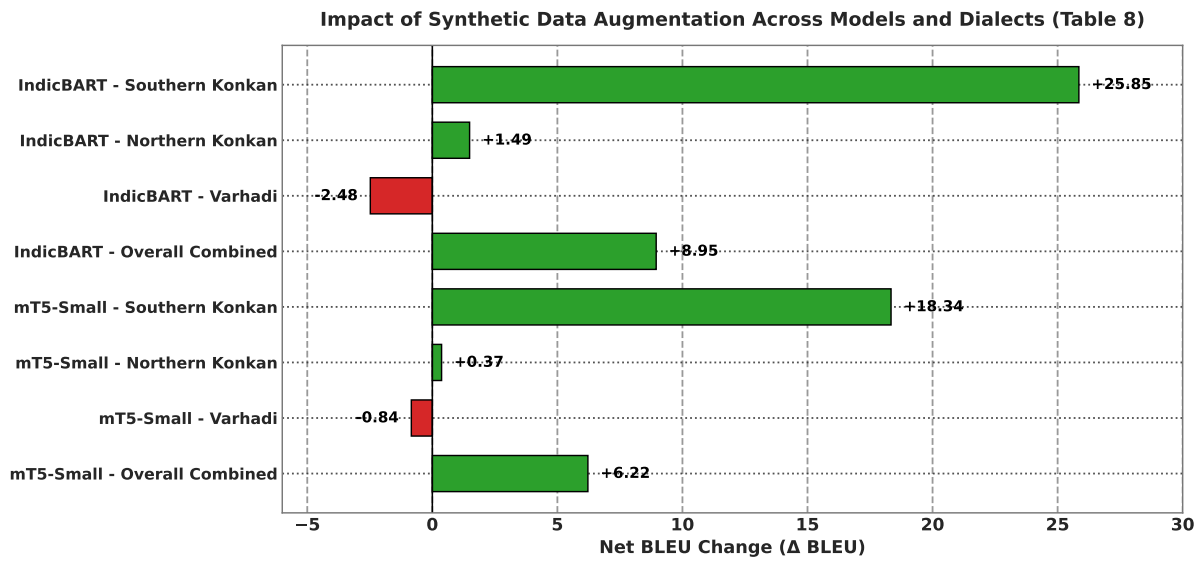


Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

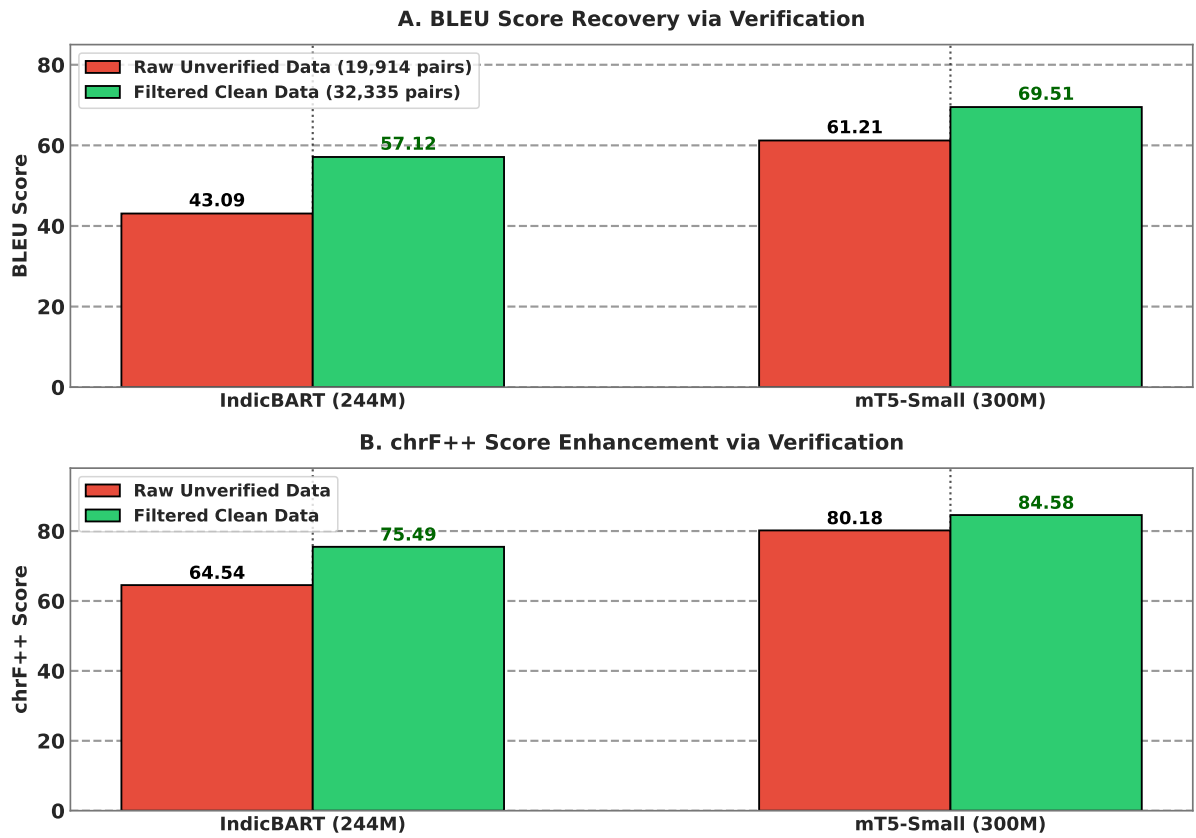


Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

5 Conclusion

This study presents a scalable, reproducible framework for multi-dialect Marathi text normalization, addressing the critical scarcity of parallel corpora for low-resource Indic varieties. By replacing expensive manual annotations with a synthetic parallel generation pipeline combining Gemma-4 translation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), we successfully expanded the clean parallel corpus from 16,163 to 32,335 verified pairs across Southern Konkan, Northern Konkan, and Varhadi dialects. Empirical evaluations demonstrate that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect dataset (+12.39 BLEU over IndicBART). Crucially, multi-tier verification prevented synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion mitigated cross-dialect parameter competition and maintained near-perfect Standard Marathi normalization (97.39 BLEU, 1.37% WER) on held-out RESPIN benchmarks.

While these findings establish a strong blueprint for Indic sub-dialect normalization, minor constraints remain: LLM-driven verification pipelines introduce computational and latency overheads, synthetic expansion can occasionally yield slight precision dilution on already high-performing dialects like Varhadi, and intrinsic text metrics should be complemented by downstream task evaluations (e.g., ASR post-processing) in future work. Overall, our verified synthetic pipeline provides a scalable foundation for expanding digital linguistic inclusion across low-resource regional dialects.

References

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, pages 5998–6008, 2017. URL <https://arxiv.org/abs/1706.03762>.
- [2] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, and et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- [3] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6282–6293, 2020. doi: 10.18653/v1/2020.acl-main.560.
- [4] Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R Rickford, Dan Jurafsky, and Sharad Goel. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689, 2020.
- [5] Marcos Zampieri, Preslav Nakov, and Yves Scherrer. Natural language processing for similar languages, varieties, and dialects: A survey. *Natural Language Engineering*, 26(6):605–637, 2020.
- [6] Verena Blaschke, Christoph Purschke, Hinrich Schütze, and Barbara Plank. What do dialect speakers want? a survey of attitudes towards language technology for german dialects. *arXiv preprint arXiv:2402.11968*, 2024. URL <https://arxiv.org/abs/2402.11968>.
- [7] Sonal Kulkarni-Joshi. Variation and change in dialects of marathi: A social-dialectological approach. In Pritha Chandra, editor, *Variation in South Asian Languages: From Macro to Micro-Differences*, pages 207–236. Springer, 2023. doi: 10.1007/978-981-99-1149-3_9.
- [8] Akriti Dhasmana, Aarohi Srivastava, and David Chiang. Dialect matters: Cross-lingual asr transfer for low-resource indic language varieties. In *Proceedings of the Thirteenth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ ACL 2026)*, pages 145–156. Association for Computational Linguistics, 2026.
- [9] Nazmuddoha Ansary, Quazi Adibur Rahman Adib, Tahsin Reasat, Asif Shahriyar Sushmit, Ahmed Imtiaz Humayun, Sazia Mehnaz, Kanij Fatema, Mohammad Mamun Or Rashid, and Farig Sadeque. Unicode normalization and grapheme parsing of indic languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 17019–17030. ELRA Language Resource Association, 2024.
- [10] Aditya Joshi, Raj Dabre, Diptesh Kanojia, Zhuang Li, Haolan Zhan, Gholamreza Haffari, and Doris Dippold. Natural language processing for dialects of a language: A survey. *ACM Computing Surveys*, 57(6):1–37, 2025. URL <https://doi.org/10.1145/3712060>.

- [11] Raj Dabre, Himani Shrotriya, Anoop Kunchukuttan, Ratish Puduppully, Mitesh M. Khapra, and Pratyush Kumar. Indicbart: A pre-trained model for indic natural language generation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1849–1863, 2022. URL <https://doi.org/10.18653/v1/2022.findings-acl.145>.
- [12] Divyanshu Kakwani, Anoop Kunchukuttan, Satish Golla, N. C. Gokul, Pushpak Bhattacharyya, Mitesh M. Khapra, and Pratyush Kumar. Indicnlp suite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for indic languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4961. Association for Computational Linguistics, 2020.
- [13] Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mt5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT 2021*, pages 483–498, 2021. URL <https://doi.org/10.18653/v1/2021.naacl-main.41>.
- [14] Aman Kumar, Manish Shrivastava, and Pushpak Bhattacharyya. Indic-dialect: Benchmark dataset for indic dialect classification and translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15201–15212. ELRA Language Resource Association, 2024.
- [15] Ona de Gibert et al. Scaling low-resource mt via synthetic data generation with llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.
- [16] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.
- [17] Saurabh Kumar, Abhayjeet Singh, Deekshitha G, Amartyaveer, Jesuraja Bandekar, Savitha Murthy, Sumit Sharma, Sandhya Badiger, Sathvik Udupa, Amala Nagireddi, Srinivasa K. M. Raghavan, Rohan Saxena, Jai Nanavati, Raoul Nanavati, Janani Sridharan, Arjun Mehta, Ashish K. S. Khuraishi, Sai Praneeth Reddy Mora, Prashanthi Venkataramakrishnan, Gauri Date, Karthika P, and Prasanta Kumar Ghosh. RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine indian languages. In *Advances in Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*, pages 1–13, 2025.
- [18] Yves Scherrer and Nikola Ljubešić. Normalising swiss german dialect text using character-level machine translation. In *Proceedings of the 3rd Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2016)*, pages 117–126, 2016.
- [19] Mathilde Lusetti, Tsetsu Ruzsics, Tanja Samardžić, and Yves Scherrer. Impact of target language models on swiss german dialect normalization. In *Proceedings of the 5th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2018)*, pages 133–141, 2018.
- [20] Mihael Kresic and Noorhan Abbas. Normalizing swiss german dialects with the power of large language models. *Procedia Computer Science*, 244:287–295, 2024.
- [21] Niko Partanen, Mika Härmäläinen, and Khalid Alnajjar. Dialect text normalization to normative standard finnish. In *Proceedings of the 5th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2019)*, pages 141–146. Association for Computational Linguistics, 2019.
- [22] Mika Härmäläinen, Niko Partanen, and Khalid Alnajjar. Normalizing swedish dialects spoken in finland. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2020)*, pages 55–62, 2020.
- [23] Mika Härmäläinen, Niko Partanen, and Jack Rueter. Generating dialectal finnish for dialect normalization of estonian dialects. In *Proceedings of the 9th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2022)*, pages 71–79, 2022.
- [24] Olli Kuperinen, Aleksandra Miletic, and Yves Scherrer. Dialect-to-standard normalization: A large-scale multilingual evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13814–13828. Association for Computational Linguistics, 2023.
- [25] Kaori Abe, Kentaro Inui, and Yuji Matsumoto. Multi-dialect neural machine translation for japanese dialects. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2105–2116, 2018.
- [26] Yuan Zhao and Eleanor Chodroff. Acoustic-phonetic variation in mandarin dialect corpora. *Journal of Phonetics*, 94:101172, 2022.
- [27] Nizar Habash, Mona Diab, and Owen Rambow. Conventional orthography for dialectal arabic. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC '12)*, pages 711–718, 2012.
- [28] Houda Bouamor, Nizar Habash, Mohammad Salameh, et al. The madar parallel corpus of arabic dialects. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC 2018)*, pages 3375–3381, 2018.
- [29] Muhammad Abdul-Mageed, Chunsheng Zhang, AbdelRahim Elmadany, and Ebrahim Mohamed Nagoudi. Nadi 2023: The fourth nuanced arabic dialect identification and translation shared task. In *Proceedings of the Arabic NLP Conference (ArabicNLP @ EMNLP 2023)*, pages 512–528, 2023.

- [30] Fadhl Eryani, Nizar Habash, Faisal Al-Shargi, et al. A normalization corpus for five arabic dialects. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*, pages 5750–5757, 2020.
- [31] Antonios Dimakis, John Pavlopoulos, and Antonios Anastasopoulos. Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714. Association for Computational Linguistics, 2025.
- [32] Raviraj Joshi, Parth Goel, and Rucha Joshi. L3cube-mahanlp: Marathi natural language processing datasets, models, and library. In *Proceedings of the WILDRE-6 Workshop, LREC 2022*, pages 10–17, 2022. URL <https://aclanthology.org/2022.wildre-1.2>.
- [33] Vineel Pratap, Ankit Saraf, and Pratyush Kumar. Wfst-guided neural text normalization for indic languages. In *Proceedings of Interspeech 2024*, pages 3120–3124, 2024.
- [34] Rahesha Mulla and B. Suresh Kumar. Text-independent automatic dialect recognition of marathi language using spectro-temporal characteristics of voice. *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 10(2s):313–321, 2022. doi: 10.17762/ijritcc.v10i2s.5949.
- [35] Poonam S. Pawar, Jatinderkumar R. Saini, and Shraddha Vaidya. Automatic marathi dialect classification in noisy and non-noisy environments. *Procedia Computer Science*, 282:288–303, 2026.
- [36] Gaurav Gaikwad, Deepak Panchal, Omprasad Deshmukh, and Sonali Patil. Marathi dialect to pure marathi using rag & ai. *International Research Journal of Engineering and Technology (IRJET)*, 12(11):787–791, 2025. ISSN 2395-0056.
- [37] V. A. Vyavhare, S. N. Divekar, Harshada Pawar, Karan Kale, and Subhash Jagtap. Marathi dialect translator. *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, 13(3):254–261, 2026. doi: 10.32628/IJSRSET2613331.
- [38] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions, 2022.
- [39] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*, 2023.
- [40] Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Sahaj Agarwal, Hamid Palangi, and Ahmed Awadallah. Orca: Progressive learning from complex explanation traces of gpt-4, 2023.
- [41] Cheng-Yu Hsieh et al. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*, 2023.
- [42] Branislav Pecher et al. Better as generators than classifiers: Leveraging llms and synthetic data for low-resource multilingual classification. In *Findings of the Association for Computational Linguistics: EACL 2024*, 2024.
- [43] Google DeepMind Gemma Team. Gemma 4: Open language models. *arXiv preprint arXiv:2408.00118*, 2024. URL <https://arxiv.org/abs/2408.00118>.
- [44] Matt Post. A call for clarity in reporting bleu scores. In *Proceedings of WMT 2018*, pages 186–191, 2018. URL <https://doi.org/10.18653/v1/W18-6319>.